An Integrated Framework for Geospatial Content Extraction and Semantic Summarization from PDF Documents

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Aditya Santosh Siras, Naresh Nagamalle, Aravinda Kumar P
Department of CSE, SSGMCE, Shegaon; National Remote Sensing Centre (ISRO), Hyderabad

Author Portfolio

I. Introduction

- The "Data Trap" Problem: Government agencies and planning bodies generate millions of unstructured PDF documents (project reports, surveys, gazettes) annually. These documents contain critical location-specific data (e.g., “Bridge construction progress in Dibrugarh, Assam”) that remains "trapped" in text format, effectively invisible to geospatial analysis tools.
- The Manual Bottleneck: Extracting this spatial intelligence currently requires manual reading and data entry, which is prohibitively time-consuming and error-prone. This creates a disconnect between Natural Language (reports) and Spatial Data Infrastructures (maps).
- The Proposed Solution: We present a novel, privacy-focused Integrated Framework that automates the pipeline from "Raw PDF" to "Interactive Map.“
- Key Innovations:
  - Fully Offline Execution: Unlike cloud-based APIs, our system runs locally using Open Source LLMs (Mistral-7B), ensuring strict data privacy for sensitive government files.
  - Indian Context Awareness: Specifically designed to handle the hierarchical administrative structure of India (State to District to Tehsil).
  - Democratization: Enables non-technical policymakers to visualize complex textual reports on ISRO Bhuvan maps without requiring GIS expertise.

II. Literature Review

- Geospatial NER: Standard models fail on Indian administrative hierarchies. We fine-tuned a spaCy transformer on Indian Census & LGD data to resolve ambiguities (e.g., distinguishing a District from a Town).
- RAG for PDFs: Standard RAG systems struggle with complex PDF layouts (tables/columns). Our framework uses OCR & structural parsing before vectorization to ensure accurate, context-aware summaries.
- Research Gap: Existing tools are either cloud-dependent or lack spatial-semantic integration. This work bridges NLP and GIS in a fully offline, secure environment.

III. Problem Statement

- The Unstructured Data Silo: Massive volumes of geospatial intelligence are locked inside unstructured PDF narratives, complex multi-column tables, and scanned notices, rendering them invisible to standard GIS software.
- Inefficient Manual Workflows: Extracting location-specific insights (e.g., “Project X delayed in District Y”) currently necessitates tedious manual reading and data entry, which is unscalable and highly prone to human error.
- Linguistic & Spatial Ambiguity: Generic global AI models struggle with the Indian context. They often fail to distinguish between identical place names across states (e.g., Aurangabad) or understand administrative hierarchies (Tehsil vs. District).
- Lack of Semantic Context: Merely plotting a coordinate is insufficient for high-level planning. Decision-makers lack the contextual "why“ such as funding bottlenecks or environmental clearances linked directly to the map location.
- Data Sovereignty & Security: Government bodies deal with sensitive data that cannot be offloaded to public cloud APIs (like OpenAI or Google Maps). There is a critical gap for secure, air-gapped, and offline-first AI solutions.
- Fragmented Tooling: Currently, no single open-source framework seamlessly integrates Optical Character Recognition (OCR), NLP, and GIS visualisation specifically tailored for the Indian public sector.

IV. Methodology

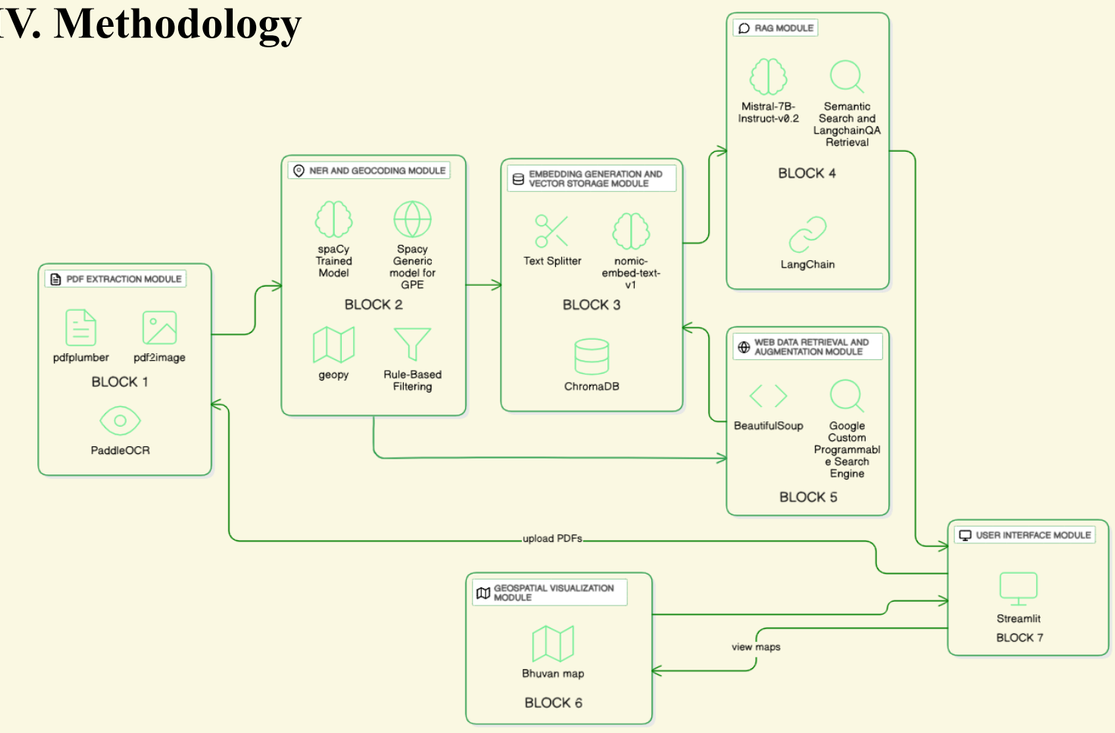


Fig. 1. Architecture of the framework.

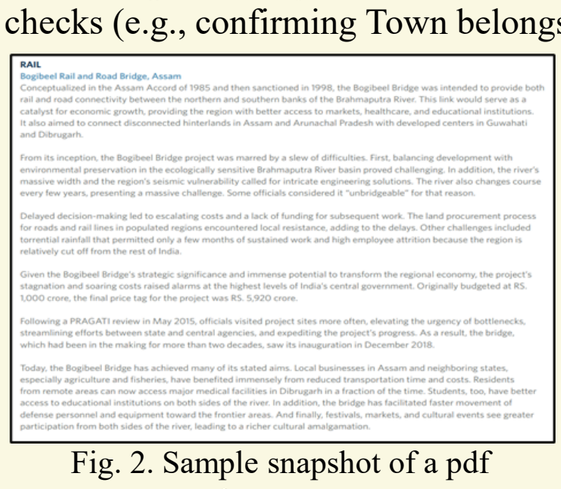
- Block 1: Ingestion: Hybrid pipeline using pdfplumber (digital text/tables) and PaddleOCR (scanned images) to digitize diverse government reports.
- Block 2: Spatial Intelligence: Custom spaCy NER model fine-tuned on Indian Census data. Uses GeoPy and hierarchical validation (Town in District) to prevent false positives.
- Block 3: Vector Storage: Splits text into semantic chunks; encodes them using nomic-embed-text-v1 and stores them in ChromaDB.
- Block 4: RAG Engine: A local, air-gapped Mistral-7B LLM retrieves relevant vector chunks to generate context-aware summaries ("Urgent Issues") offline.
- Block 5: Web Augmentation (Optional): Integrates Google Custom Search to verify local data against trusted online sources (if connectivity permits).
- Block 6: Visualization: Maps extracted entities onto ISRO Bhuvan WMS (Satellite/Base layers) using OpenLayers for interactive analysis.
- Block 7: User Interface: A low-code Streamlit frontend enabling non-technical users to upload PDFs and view insights without GIS training.

V. Results and Analyses

Performance Metrics:

- Validation F1: 0.906 (High accuracy on training data).
- Real-World Precision: 1.000 (Zero false positives; system prioritizes mapping accuracy over quantity).

Geocoding Accuracy: Achieved >95% success by using hierarchical validation checks (e.g., confirming Town belongs to the correct District).


RAIL
Bogibeel Rail and Road Bridge, Assam
Conceptualized in the Assam Accord of 1985 and then sanctioned in 1998, the Bogibeel Bridge was intended to provide both rail and road connectivity between the northern and southern banks of the Brahmaputra River. This link would serve as a catalyst for economic growth, providing the region with better access to markets, healthcare, and educational institutions. It also aimed to connect disconnected hinterlands in Assam and Arunachal Pradesh with developed centers in Guwahati and Dibrugarh.

From its inception, the Bogibeel Bridge project was marred by a slew of difficulties. First, balancing development with environmental preservation in the ecologically sensitive Brahmaputra River basin proved challenging. In addition, the river's massive width and the region's seismic vulnerability called for intricate engineering solutions. The river also changes course every few years, presenting a massive challenge. Some officials considered it "unbridgeable" for that reason.

Delayed decision-making led to escalating costs and a lack of funding for subsequent work. The land procurement process for roads and rail lines in populated regions encountered local resistance, adding to the delays. Other challenges included torrential rainfall that permitted only a few months of sustained work and high employee attrition because the region is relatively cut off from the rest of India.

Given the Bogibeel Bridge's strategic significance and immense potential to transform the regional economy, the project's stagnation and soaring costs raised alarms at the highest levels of India's central government. Originally budgeted at RS. 1,000 crore, the final price tag for the project was RS. 5,920 crore.

Following a PRAGATI review in May 2015, officials visited project sites more often, elevating the urgency of bottlenecks, streamlining efforts between state and central agencies, and expediting the project's progress. As a result, the bridge, which had been in the making for more than two decades, saw its inauguration in December 2018.

Today, the Bogibeel Bridge has achieved many of its stated aims. Local businesses in Assam and neighboring states, especially agriculture and fisheries, have benefited immensely from reduced transportation time and costs. Residents from remote areas can now access major medical facilities in Dibrugarh in a fraction of the time. Students, too, have better access to educational institutions on both sides of the river. In addition, the bridge has facilitated faster movement of defense personnel and equipment toward the frontier areas. And finally, festivals, markets, and cultural events see greater participation from both sides of the river, leading to a richer cultural amalgamation.

Fig. 2. Sample snapshot of a pdf document.

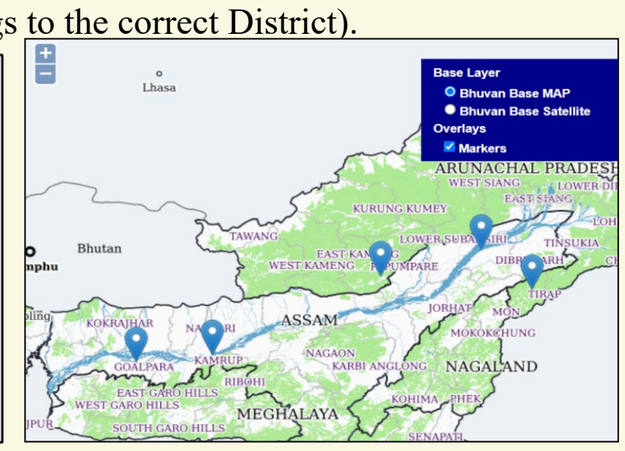


Fig. 3. Extracted entities visualized on ISRO Bhuvan WMS Base Layer.

Case Study (Bogibeel Bridge):

- Spatial Output: Accurately mapped key locations: Dibrugarh, Guwahati, and Brahmaputra River.
- RAG Insights: Automatically extracted "Urgent Issues" such as land procurement delays and cost escalation (from ₹1,000 Cr to ₹5,920 Cr).

Efficiency: The entire pipeline runs seamlessly on a standard mid-range laptop (16GB RAM), proving viability for offline field deployment.

VI. Conclusion & Future Scope

- **Conclusion**: Developed a privacy-focused, offline framework that democratizes geospatial analysis with 1.000 Precision. The solution is lightweight, open-source, and easily deployable on standard hardware.
- **Future Scope**: Extending capabilities to extract complex geometries (roads/polygons) and enabling direct GIS interoperability via GeoJSON/Shapefile exports.

References

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[2] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," arXiv:2005.11401, 2020.

Digital Copy & Code

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